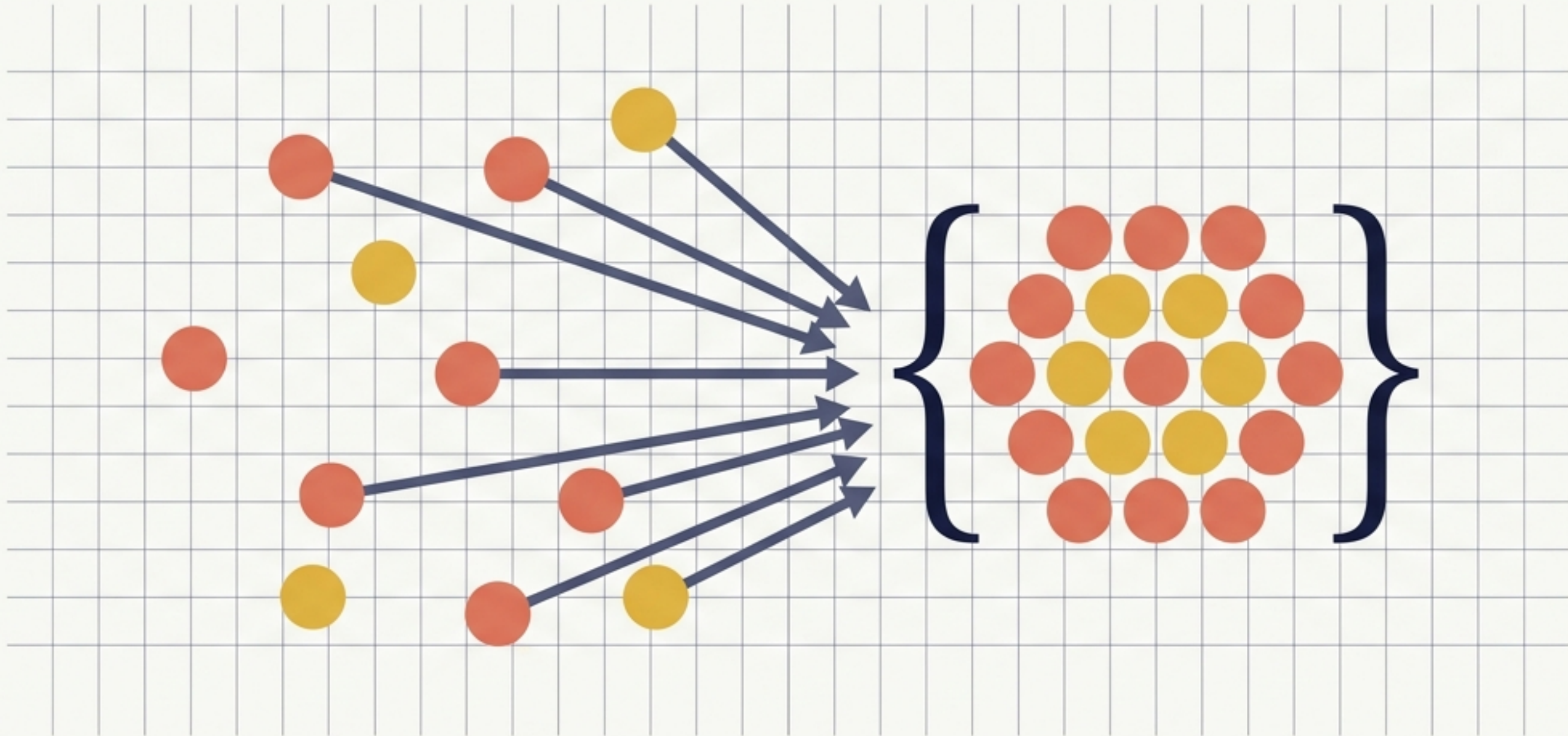


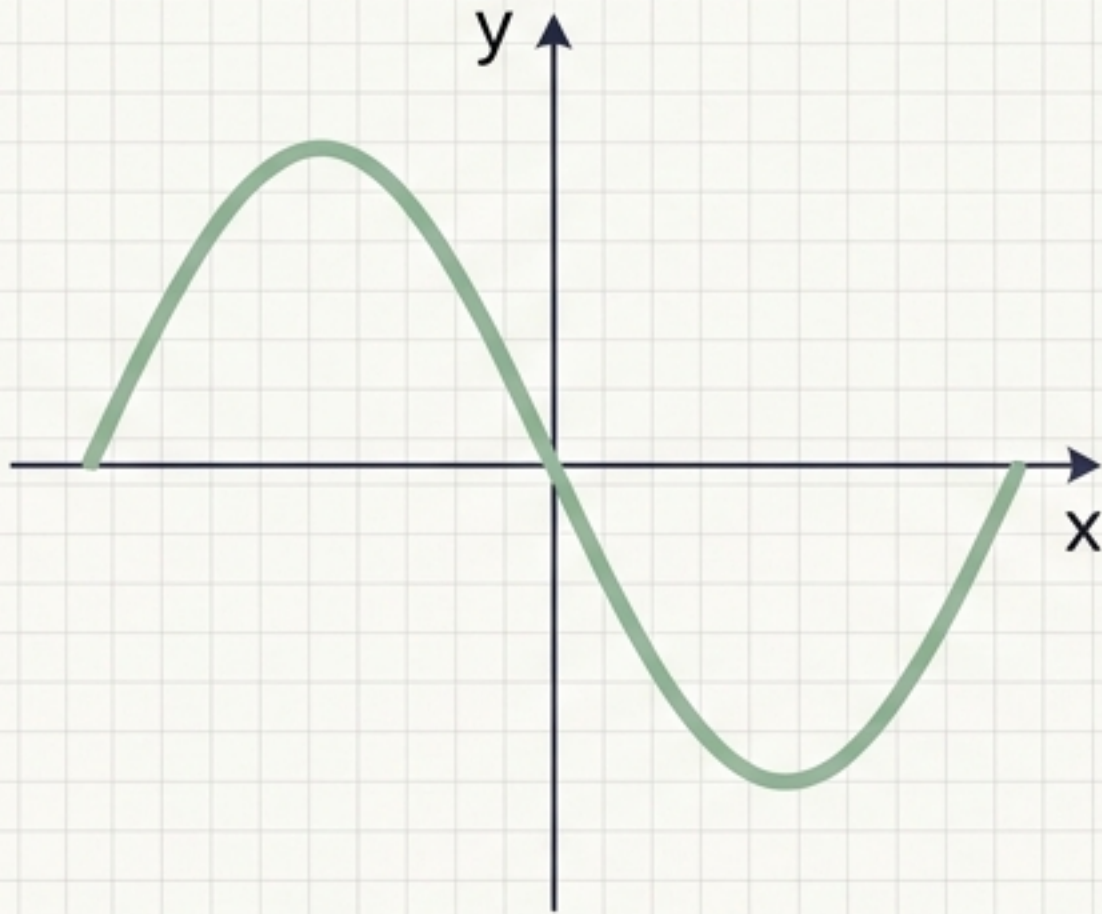


COMP 102: Discrete Mathematics

A Visual Guide to Set Theory

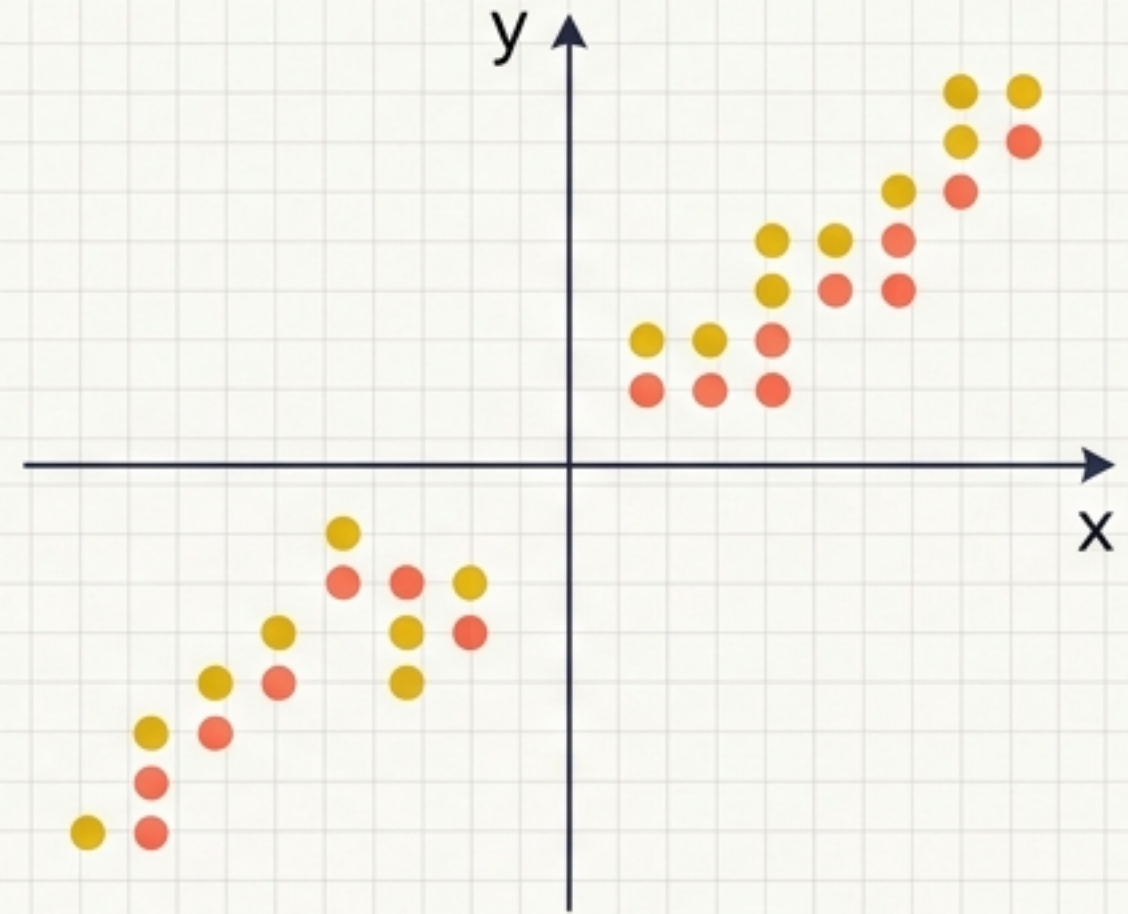
The discrete building blocks of computer science.

The Big Picture: Continuous vs. Discrete



Continuous Mathematics

Smooth quantities, infinite points between points. (e.g., Calculus)



Discrete Mathematics

Separate, distinct values, countable objects. The absolute foundation for algorithms, data structures, and cryptography.

Introduction to Sets

A set is an unordered collection of distinct objects (elements/members).



Key Rule 1:
Order does not matter.

Key Rule 2:
Repeating an element changes nothing.

Origin Note: Introduced by German mathematician Georg Cantor, Set Theory is the bedrock of graph theory and finite state machines.

The Language of Sets: Membership

Let $S = \{1, 2, 3\}$



$2 \in S$

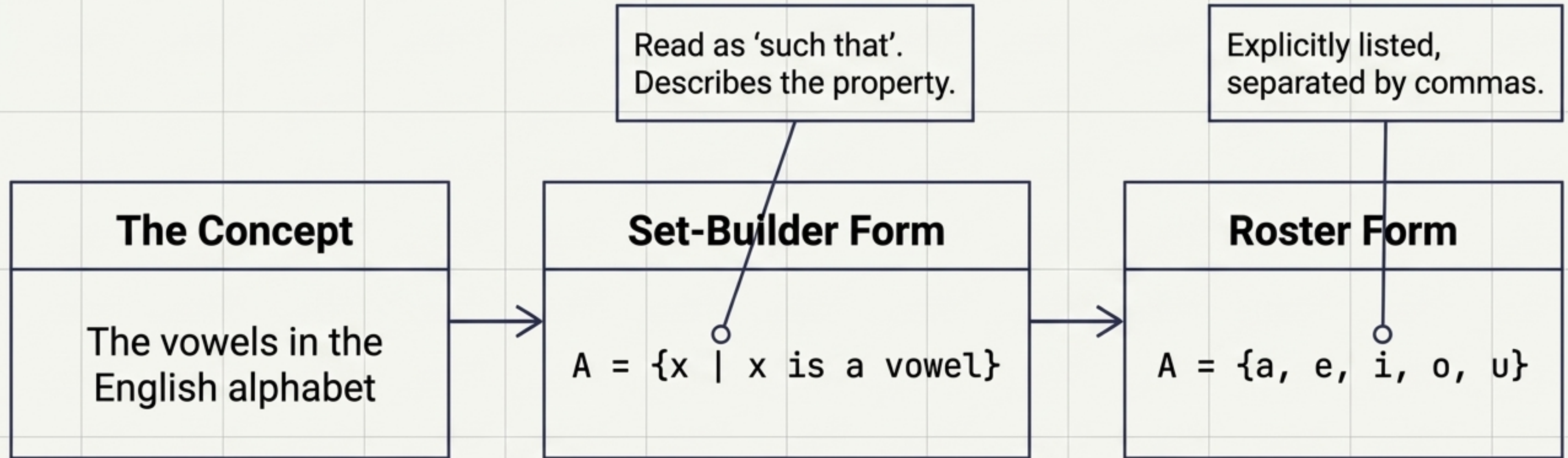
Read: 2 belongs to S



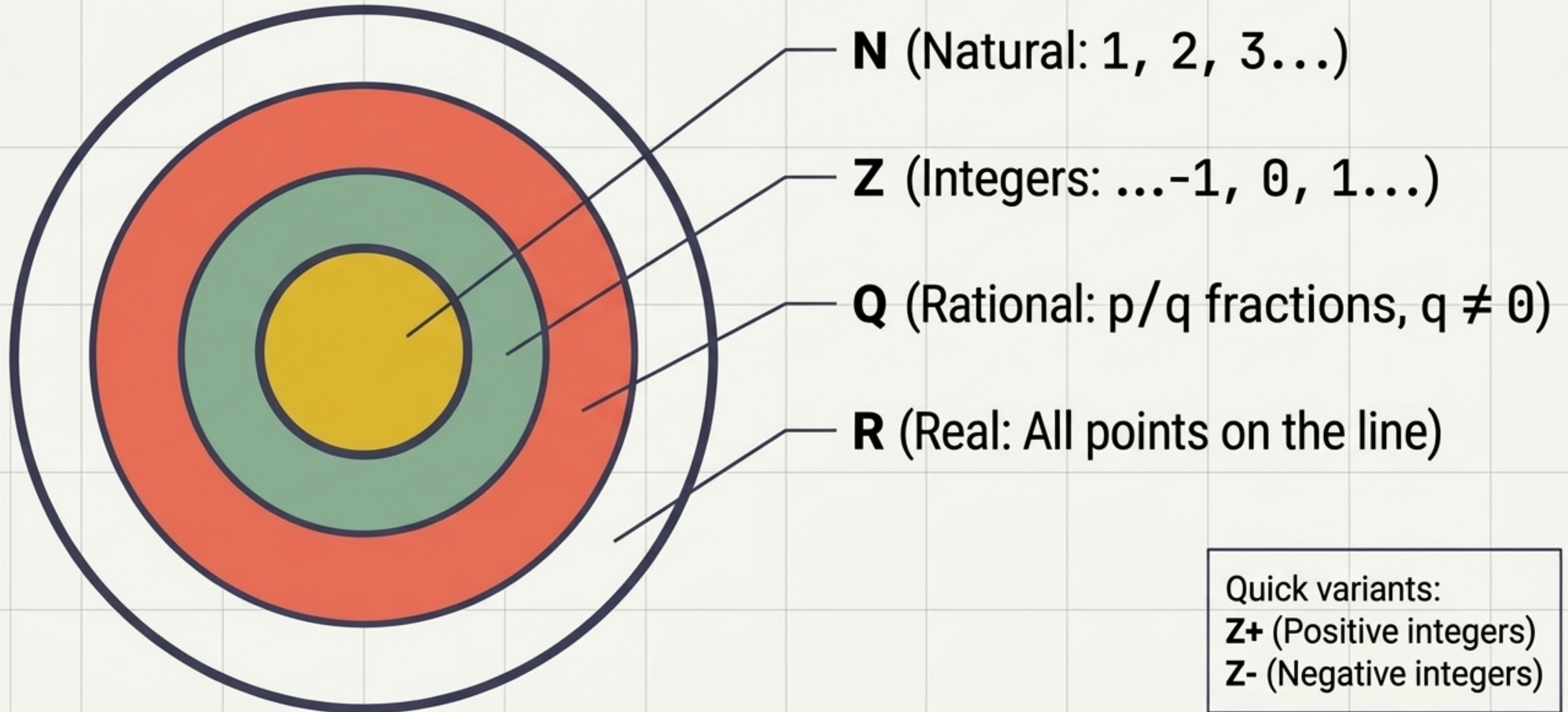
$5 \notin S$

Read: 5 does not belong to S

Describing Sets: The Translation Engine



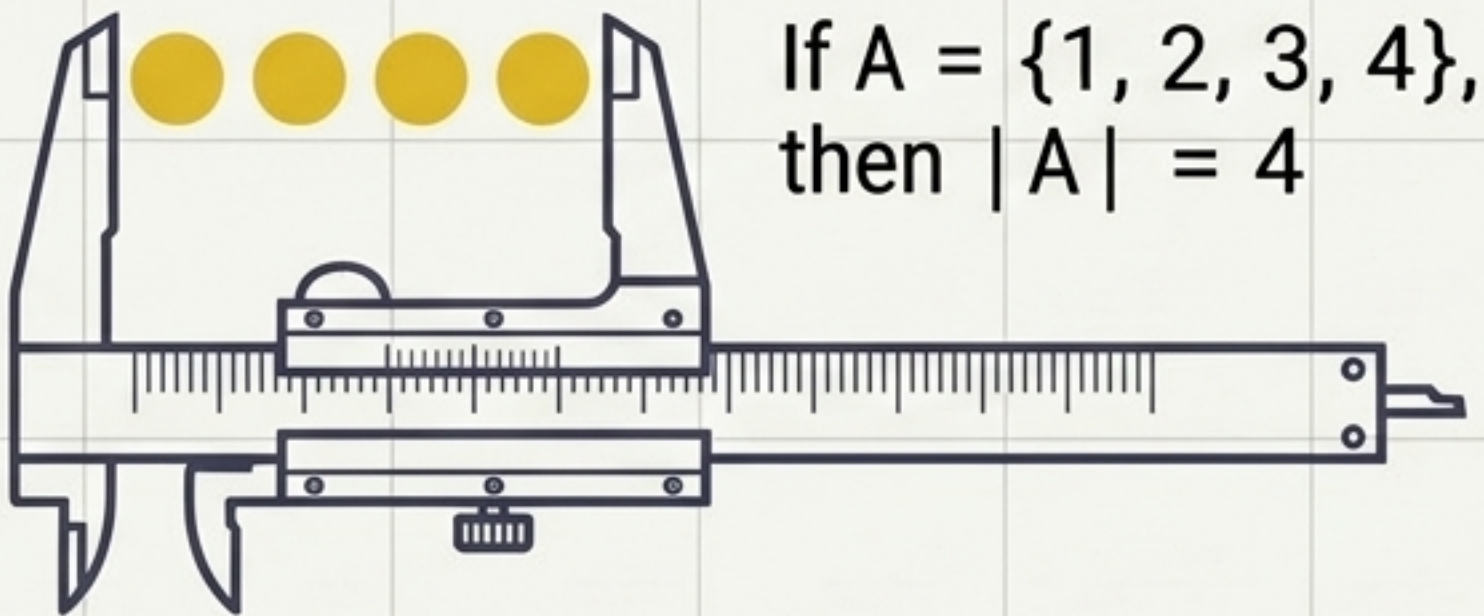
Special Sets of Numbers (Nested Universes)



Cardinality: Measuring Sets

Cardinality (written as $|S|$) is the absolute number of elements in Set S .

Example A (Countable)



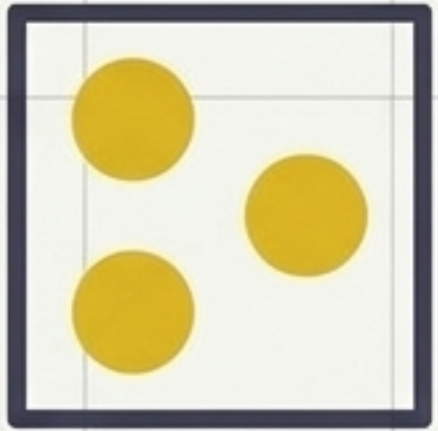
Example B (Infinite)



For sets with infinitely many elements, we write $|S| = \infty$

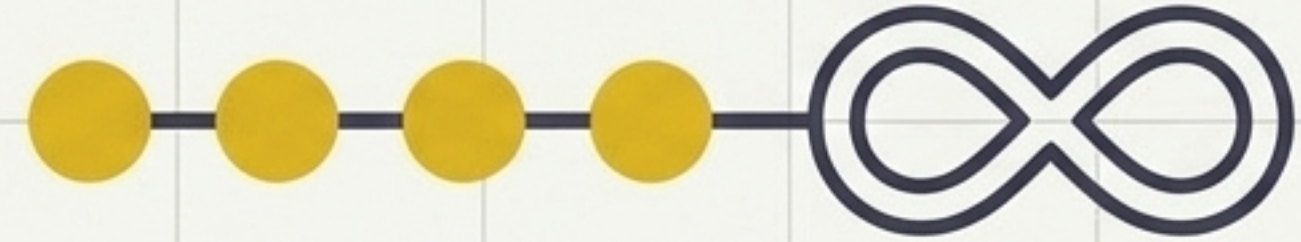
The Taxonomy of Sets

Finite Set



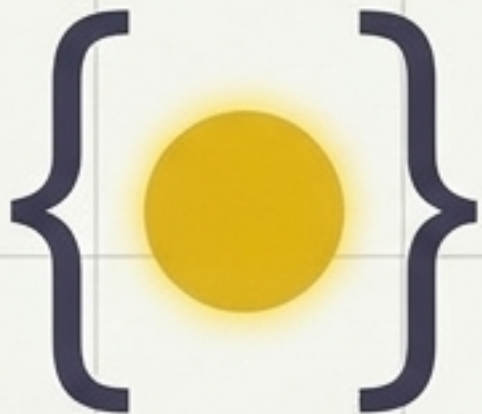
Definite, limited, and countable elements.

Infinite Set



Never-ending elements.
e.g., Set of all Natural numbers ($\mathbb{N}$).

Singleton (Unit) Set



Exactly one element.
e.g., $\{5\}$.

Empty (Null) Set



No elements.
Denoted as $\emptyset$ or $\{ \}$.
Cardinality is 0.

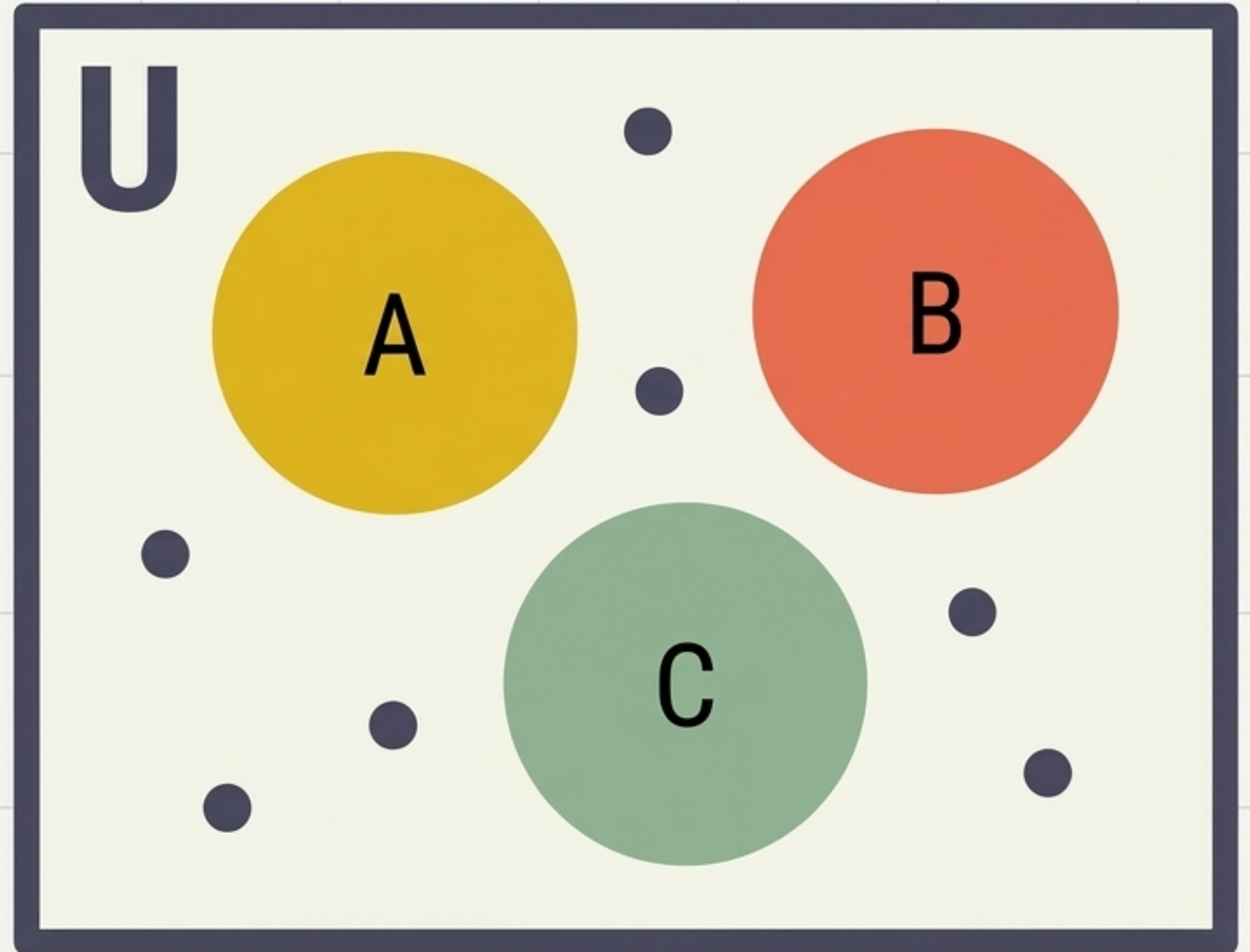
The Universal Set

The “**Universe of Discourse.**”

The set of all objects under consideration in a specific context.

Rule: Every other set in that discussion is a subset of U.

Example: If studying wildlife,
 $U = \{\text{All animals on Earth}\}.$



Comparison Matrix: Equal vs. Equivalent

EQUAL SETS

Contain exactly the same elements.

$$\begin{array}{l} A = \{1, 2, 3\} \\ B = \{3, 1, 2\} \end{array}$$

$$A = B$$

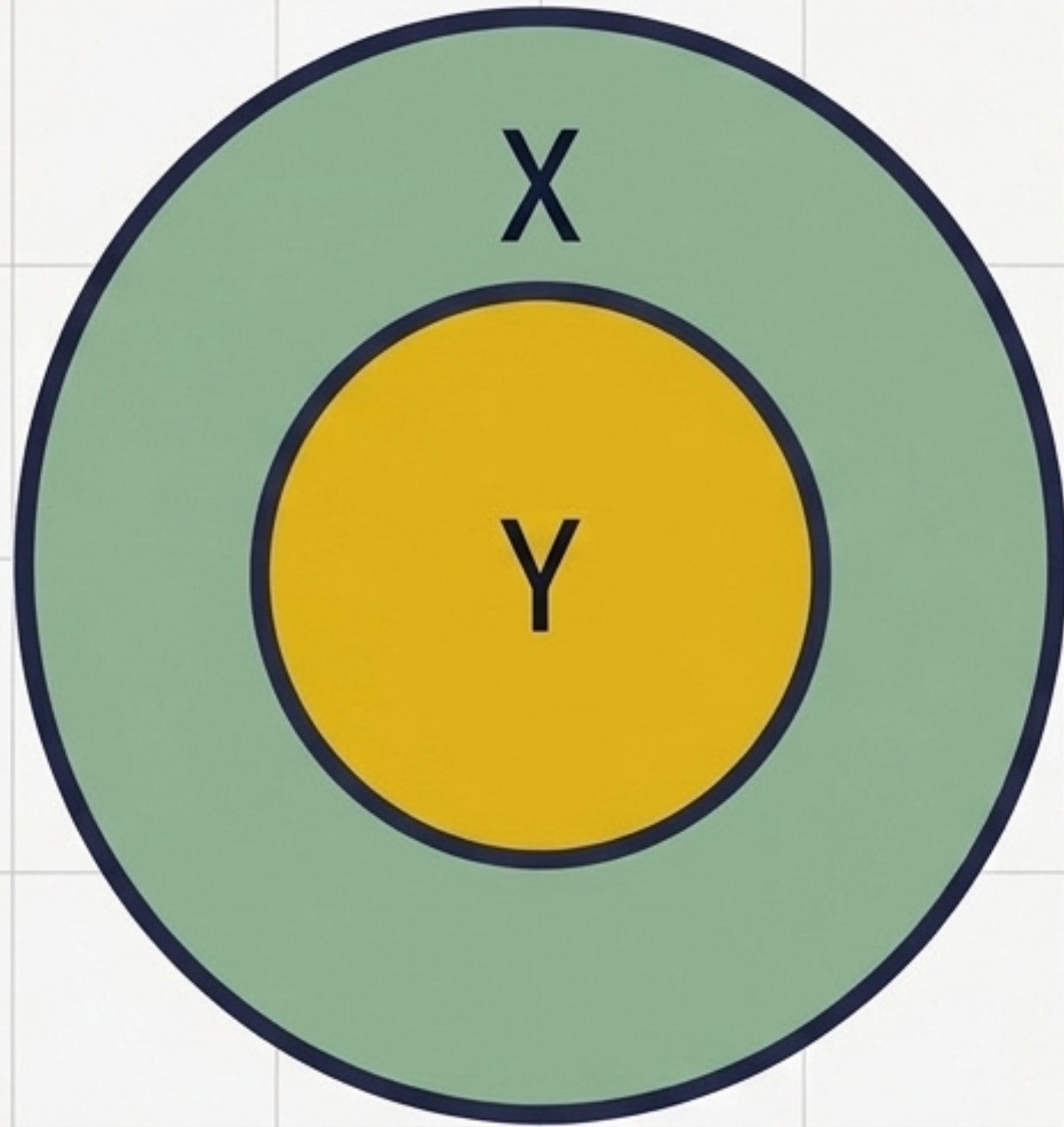
EQUIVALENT SETS

Have the same number of elements (same cardinality), but elements can differ. Requires a 1-to-1 bijection.

$$\begin{array}{l} A = \{1, 2, 3\} \\ B = \{a, b, c\} \end{array}$$

$$|A| = |B|$$

Relationships: Subsets vs. Proper Subsets



$$Y \subseteq X$$

Every element of Y is also in X.
(Note: A set can be a subset of itself, $A \subseteq A$).



$$Y \subsetneq X$$

Every element of Y is in X,
BUT $Y \neq X$. X must contain at
least one element not found in Y.

Golden Rule Callout: The empty set ($\emptyset$) is a subset of every set.

Synthesis: The Set Theory Cheat Sheet

$\{ \}$

Set (unordered
distinct objects)

$\in / \notin$

Membership
(in / not in)

$|S|$

Cardinality
(size of set)

$\emptyset$

Empty Set
(zero elements)

U

Universal Set
(all objects in context)

$\subseteq$

Subset
(all elements included)

$\subset$

Proper Subset
(subset, but strictly smaller)

Master these definitions and differences—they are the structural foundation of computer science logic.